

Ibad ullah
25K-6544

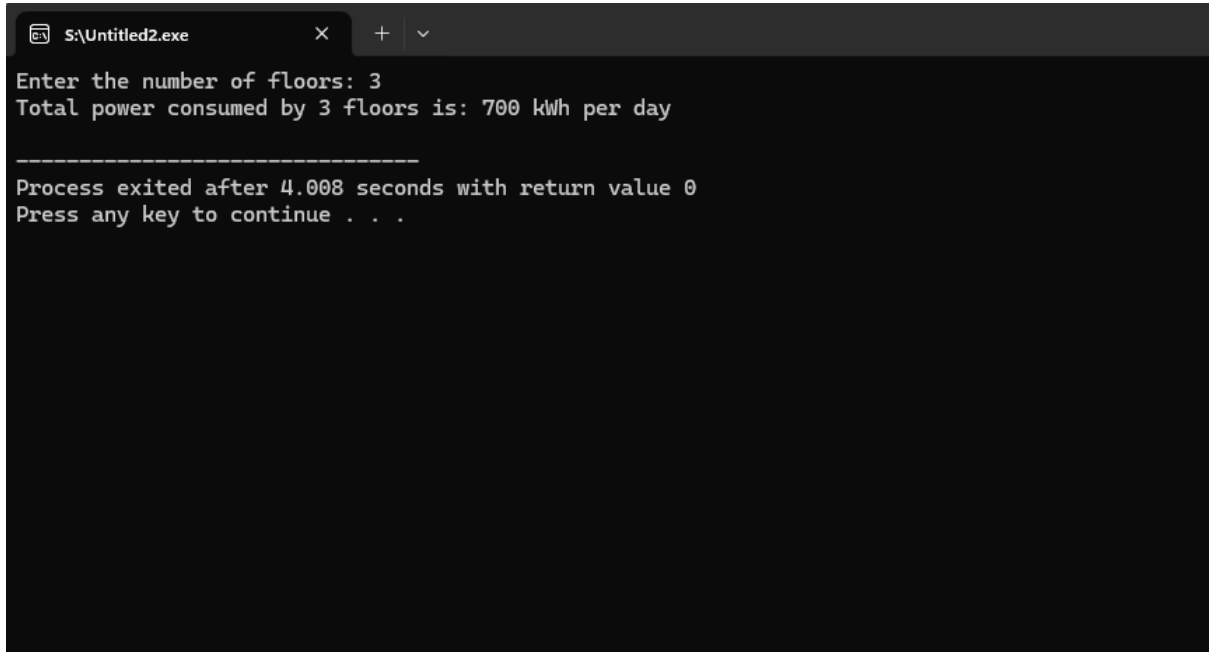
TASK 1:

```
#include <stdio.h>
```

```
int totalPower(int floors) {  
    if (floors <= 1) {  
        return 100;  
    }  
    int currentPower = 100;  
    for (int i = 1; i < floors; i++) {  
        currentPower = currentPower * 2;  
    }  
    return currentPower + totalPower(floors - 1);  
}
```

```
int main() {  
    int floors;  
    int total;  
    printf("Enter the number of floors: ");  
    scanf("%d", &floors);  
  
    if (floors < 1) {  
        printf("Invalid input. Floors must be 1 or greater.\n");  
        return 0;  
    }  
    total = totalPower(floors);  
    printf("Total power consumed by %d floors is: %d kWh per day\n", floors, total);  
    return 0;  
}
```

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```
S:\Untitled2.exe X + v
Enter the number of floors: 3
Total power consumed by 3 floors is: 700 kWh per day

-----
Process exited after 4.008 seconds with return value 0
Press any key to continue . . .
```

TASK 2:

```
#include <stdio.h>
```

```
int countUpper(char *str, int *digitCount) {
```

```
    if (str[0] == '\0') {
```

```
        return 0;
```

```
    }
```

```
    int isUpper = 0;
```

```
    if (str[0] >= 'A' && str[0] <= 'Z') {
```

```
        isUpper = 1;
```

```
    }
```

```
    if (str[0] >= '0' && str[0] <= '9') {
```

```
        *digitCount = *digitCount + 1;
```

```
    }
```

```
    return isUpper + countUpper(str + 1, digitCount);
```

```
}
```

```
int main() {
```

```
    char password[100];
```

```
    int digits = 0;
```

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```
int uppers;

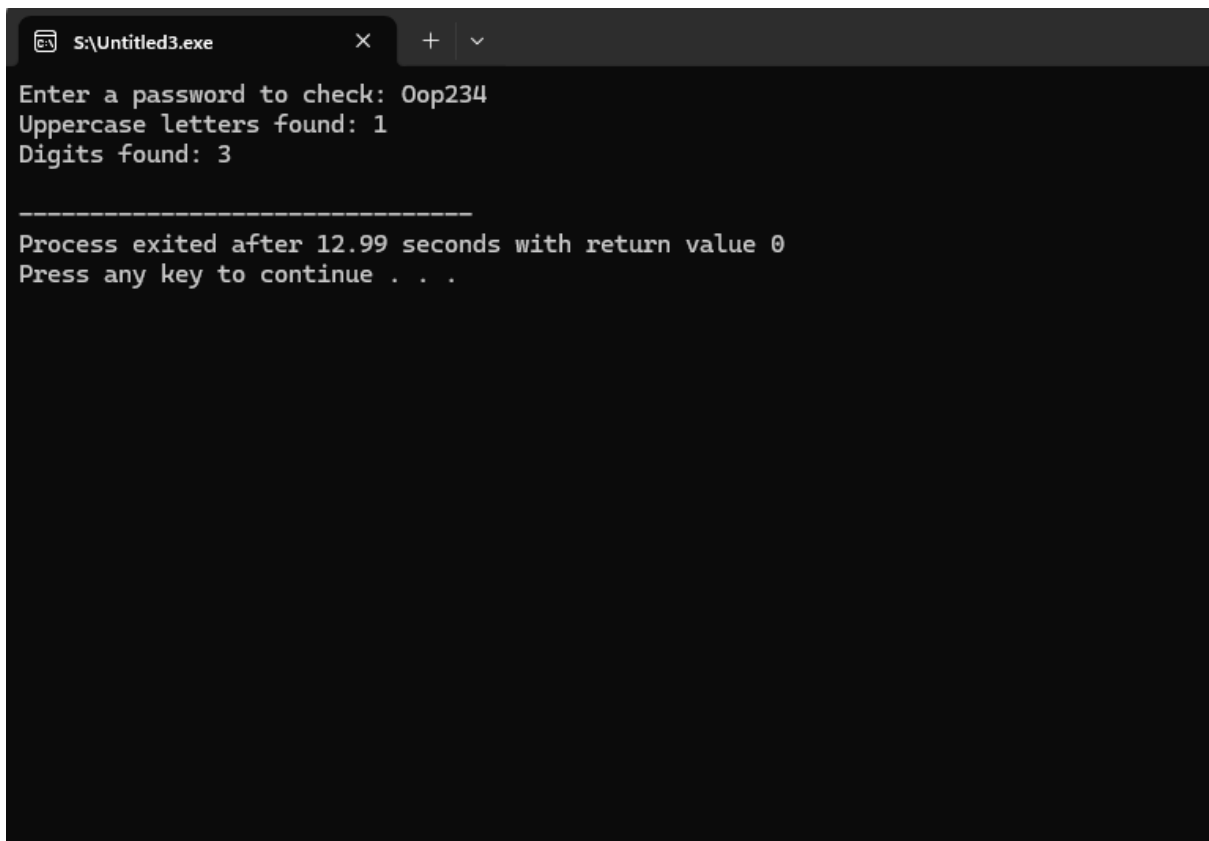
printf("Enter a password to check: ");
scanf("%s", password);

uppers = countUpper(password, &digits);

printf("Uppercase letters found: %d\n", uppers);

printf("Digits found: %d\n", digits);

return 0;
}
```



```
S:\Untitled3.exe
Enter a password to check: Oop234
Uppercase letters found: 1
Digits found: 3

-----
Process exited after 12.99 seconds with return value 0
Press any key to continue . . .
```

TASK 3:

```
#include <stdio.h>

int countWays(int n) {
    if (n < 0) {
        return 0;
    }
    if (n == 0 || n == 1) {
        return 1;
    }
}
```

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```
}
```

```
return countWays(n - 1) + countWays(n - 2);
```

```
}
```

```
void printPaths(int n, int path[], int pathLen) {
```

```
    if (n < 0) {
```

```
        return;
```

```
    }
```

```
    if (n == 0) {
```

```
        for (int i = 0; i < pathLen; i++) {
```

```
            printf("%d", path[i]);
```

```
            if (i < pathLen - 1) {
```

```
                printf("+");
```

```
            }
```

```
        }
```

```
        printf("\n");
```

```
        return;
```

```
    }
```

```
    path[pathLen] = 1;
```

```
    printPaths(n - 1, path, pathLen + 1);
```

```
    path[pathLen] = 2;
```

```
    printPaths(n - 2, path, pathLen + 1);
```

```
}
```

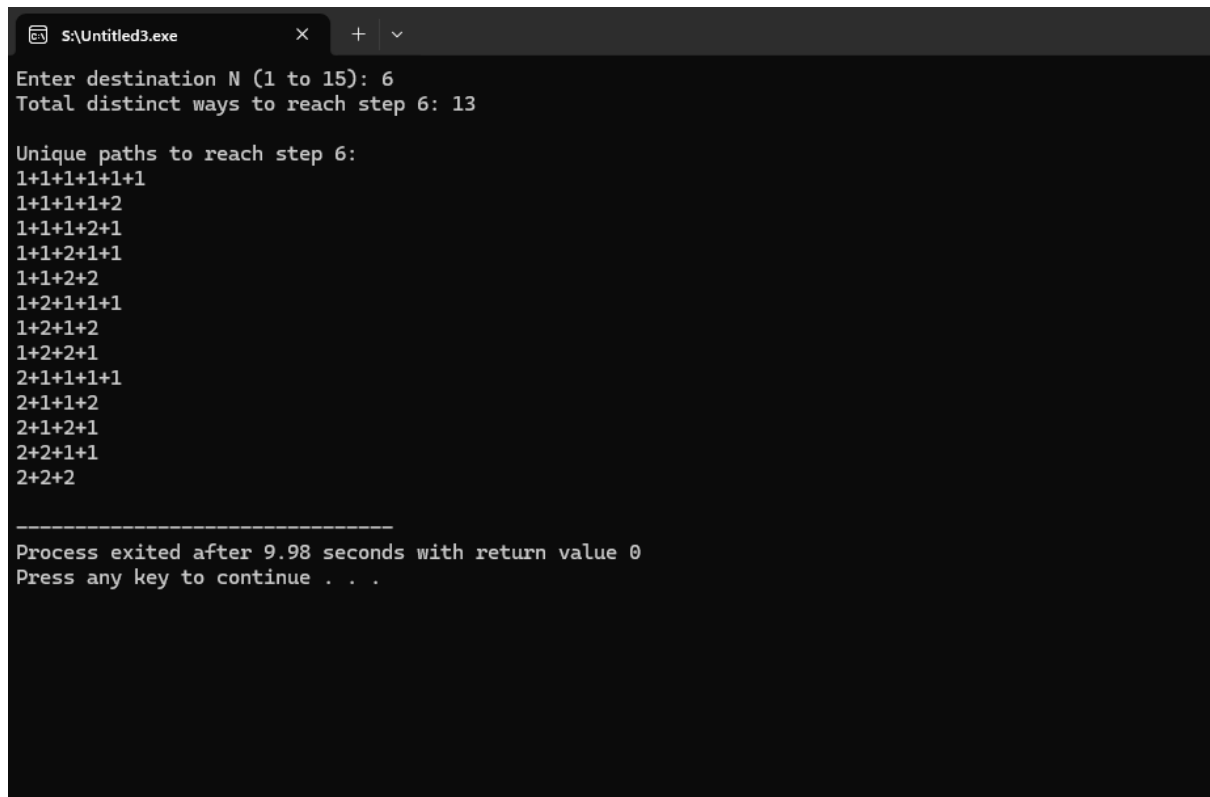
```
int main() {
```

```
    int n;
```

```
    int pathTracker[20];
```

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```
printf("Enter destination N (1 to 15): ");  
scanf("%d", &n);  
  
if (n < 1 || n > 15) {  
    printf("Invalid input. Please enter between 1 and 15.\n");  
    return 0;  
}  
  
int total = countWays(n);  
printf("Total distinct ways to reach step %d: %d\n\n", n, total);  
  
printf("Unique paths to reach step %d:\n", n);  
printPaths(n, pathTracker, 0);  
  
return 0;  
}
```



```
S:\Untitled3.exe
Enter destination N (1 to 15): 6
Total distinct ways to reach step 6: 13

Unique paths to reach step 6:
1+1+1+1+1+1
1+1+1+1+2
1+1+1+2+1
1+1+2+1+1
1+1+2+2
1+2+1+1+1
1+2+1+2
1+2+2+1
2+1+1+1+1
2+1+1+2
2+1+2+1
2+2+1+1
2+2+2

-----
Process exited after 9.98 seconds with return value 0
Press any key to continue . . .
```

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TASK 4:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
struct Patient {
```

```
    char name[50];
```

```
    int age;
```

```
    char bloodType[5];
```

```
    int patientID;
```

```
    char diagnosis[100];
```

```
};
```

```
void displayAll(struct Patient p[], int n) {
```

```
    printf("\n--- Patient Records ---\n");
```

```
    printf("%-10s | %-20s | %-5s | %-10s | %-30s\n", "ID", "Name", "Age", "Blood",  
"Diagnosis");
```

```
    printf("-----\n");
```

```
    for (int i = 0; i < n; i++) {
```

```
        printf("%-10d | %-20s | %-5d | %-10s | %-30s\n",
```

```
            p[i].patientID, p[i].name, p[i].age, p[i].bloodType, p[i].diagnosis);
```

```
    }
```

```
    printf("\n");
```

```
}
```

```
void searchByID(struct Patient p[], int n, int id) {
```

```
    for (int i = 0; i < n; i++) {
```

```
        if (p[i].patientID == id) {
```

```
            printf("\n--- Patient Found ---\n");
```

```
            printf("ID: %d\n", p[i].patientID);
```

```
            printf("Name: %s\n", p[i].name);
```

```
            printf("Age: %d\n", p[i].age);
```

```
            printf("Blood Type: %s\n", p[i].bloodType);
```

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```
        printf("Diagnosis: %s\n", p[i].diagnosis);  
        return;  
    }  
}  
printf("\nPatient not found.\n");  
}
```

```
int main() {  
    struct Patient patients[5];  
    printf("Enter Hospital Patient Data \n");  
    for (int i = 0; i < 5; i++) {  
        printf("\nPatient %d:\n", i + 1);  
  
        printf("Name: ");  
        if (i == 0) {  
            gets(patients[i].name);  
        } else {  
            getchar();  
            gets(patients[i].name);  
        }  
        printf("Age: ");  
        scanf("%d", &patients[i].age);  
        printf("Blood Type: ");  
        scanf("%s", patients[i].bloodType);  
        printf("Patient ID: ");  
        scanf("%d", &patients[i].patientID);  
        printf("Diagnosis: ");  
        getchar();  
        gets(patients[i].diagnosis);  
    }  
}
```

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```
displayAll(patients, 5);

int searchId;

printf("Enter a Patient ID to search: ");

scanf("%d", &searchId);

searchById(patients, 5, searchId);

return 0;

}
```

--- Patient Records ---				
ID	Name	Age	Blood	Diagnosis
1	humza	34	A-	VIRAL
2	uzaiifa	22	A+	flu
3	ADDIE	13	O-	FEVER
4	aad	21	O+	cough
5	liza	23	O+	headache

Enter a Patient ID to search: 003

--- Patient Found ---
ID: 3
Name: ADDIE
Age: 13
Blood Type: O-
Diagnosis: FEVER

TASK 5:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
struct Department {
```

```
    char deptCode[10];
```

```
    char deptName[50];
```

```
};
```

```
struct Course {
```

```
    char courseCode[10];
```

```
    char courseName[60];
```

```
    int creditHours;
```

```
    struct Department dept;
```

```
};
```

```
void searchByDepartment(struct Course catalog[], int size, char searchCode[]) {
```

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```
int found = 0;

printf("\n--- Courses for Department: %s ---\n", searchCode);

for (int i = 0; i < size; i++) {
    if (strcmp(catalog[i].dept.deptCode, searchCode) == 0) {
        printf("Course: %s - %s\n", catalog[i].courseCode, catalog[i].courseName);
        printf("Credits: %d\n", catalog[i].creditHours);
        printf("Dept Name: %s\n\n", catalog[i].dept.deptName);
        found = 1;
    }
}

if (found == 0) {
    printf("No courses found for this department code.\n");
}

}

int main() {
    struct Course catalog[3];
    printf("Enter Course Data \n");
    for (int i = 0; i < 3; i++) {
        printf("\nCourse %d:\n", i + 1);

        printf("Course Code: ");
        scanf("%s", catalog[i].courseCode);
        getchar();
        printf("Course Name: ");
        gets(catalog[i].courseName);

        printf("Credit Hours: ");
        scanf("%d", &catalog[i].creditHours);
```

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```
printf("Department Code: ");  
scanf("%s", catalog[i].dept.deptCode);  
getchar();  
printf("Department Name: ");  
gets(catalog[i].dept.deptName);  
}
```

```
printf("\n Complete Course Catalog \n");  
for (int i = 0; i < 3; i++) {  
    printf("Course: %s - %s (%d Credits)\n",  
           catalog[i].courseCode, catalog[i].courseName, catalog[i].creditHours);  
    printf("Department: %s (%s)\n\n",  
           catalog[i].dept.deptName, catalog[i].dept.deptCode);  
}
```

```
char searchCode[10];  
printf("Enter a Department Code to search: ");  
scanf("%s", searchCode);  
searchByDepartment(catalog, 3, searchCode);  
return 0;  
}
```

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```
S:\Untitled3.exe
Course 2:
Course Code: EE1001
Course Name: LINEAR CIRCUIT ANALYSIS
Credit Hours: 3
Department Code: ELECTRICAL ENGINEERING
Department Name:
Course 3:
Course Code: CS2001
Course Name: DATA STRUCTURES
Credit Hours: 3
Department Code: COMPUTER SCIENCE
Department Name:
Complete Course Catalog
Course: CS1002 - PROGRAMMING FUNDAMENTAL (3 Credits)
Department: SCIENCE (COMPUTER)

Course: EE1001 - LINEAR CIRCUIT ANALYSIS (3 Credits)
Department: ENGINEERING (ELECTRICALENGINEERING)

Course: CS2001 - DATA STRUCTURES (3 Credits)
Department: SCIENCE (COMPUTER)

Enter a Department Code to search: CS

--- Courses for Department: CS ---
No courses found for this department code.

-----
Process exited after 116.2 seconds with return value 0
Press any key to continue . . .
```

TASK 6:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
struct Order {
    int orderID;
    char customerName[50];
    char productName[50];
    int quantity;
    float unitPrice;
    char status[20];
};
```

```
float computeTotal(struct Order o) {
    return o.quantity * o.unitPrice;
}
```

```
void filterByStatus(struct Order orders[], int n, char *status) {
```

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```
int found = 0;

printf("\n--- Orders with Status: %s ---\n", status);

for (int i = 0; i < n; i++) {
    if (strcmp(orders[i].status, status) == 0) {
        printf("Order ID: %d | Customer: %s | Product: %s\n",
            orders[i].orderID, orders[i].customerName, orders[i].productName);
        found = 1;
    }
}

if (found == 0) {
    printf("No orders found with that status.\n");
}

}

int main() {
    struct Order orders[4];

    printf("Enter E-Commerce Order Details \n");

    for (int i = 0; i < 4; i++) {
        printf("\nOrder %d:\n", i + 1);

        printf("Order ID: ");
        scanf("%d", &orders[i].orderID);
        getchar();

        printf("Customer Name: ");
        gets(orders[i].customerName);

        printf("Product Name: ");
        gets(orders[i].productName);
```

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```
    printf("Quantity: ");
    scanf("%d", &orders[i].quantity);

    printf("Unit Price: ");
    scanf("%f", &orders[i].unitPrice);
    getchar();

    printf("Status (Pending, Shipped, Delivered): ");
    gets(orders[i].status);
}
printf("\n=== Itemized Bills ===\n");
for (int i = 0; i < 4; i++) {
    float total = computeTotal(orders[i]);
    printf("Order %d (Customer: %s) -> Total Bill: $%.2f\n",
        orders[i].orderId, orders[i].customerName, total);
}

char searchStatus[20];
printf("\nEnter a status to filter by: ");
gets(searchStatus);
filterByStatus(orders, 4, searchStatus);
return 0;
}
```

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```
Order ID: 1002
Customer Name: sara
Product Name: lip gloss
Quantity: 7
Unit Price: 850
Status (Pending, Shipped, Delivered): delivered
```

```
Order 3:
Order ID: 1003
Customer Name: vania
Product Name: chips
Quantity: 5
Unit Price: 100
Status (Pending, Shipped, Delivered): shipped
```

```
Order 4:
Order ID: 1004
Customer Name: hassan
Product Name: speaker
Quantity: 1
Unit Price: 3950
Status (Pending, Shipped, Delivered): delivered
```

=== Itemized Bills ===

```
Order 1001 (Customer: sarim) -> Total Bill: $3000.00
Order 1002 (Customer: sara) -> Total Bill: $5950.00
Order 1003 (Customer: vania) -> Total Bill: $500.00
Order 1004 (Customer: hassan) -> Total Bill: $3950.00
```

Enter a status to filter by: pending

--- Orders with Status: pending ---

Order ID: 1001 | Customer: sarim | Product: headphones